

Deep Learning-Based Transfer Learning for the Detection of Leukemia

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Abstract - Leukemia is typically diagnosed by using a microscope to examine blood and bone marrow smears as well as extremely challenging procedures like the Cytochemical test, which can be used to identify and categorize leukemia. However, these techniques are incredibly pricy, dependent, and time-consuming on the skill and cunning of the experts involved. In contrast to conventional methods, which have several drawbacks, this research suggests a classification model depends upon blood microscopic images for detecting leukemia by utilizing transfer learning. The suggested study demonstrates a computer-aided diagnosis method that uses Mobilenet V2 and Convolutional Neural Networks (CNNs) to identify leukemia images over healthy images. When it comes to using deep learning for image analysis and the comparison outcomes of the current study for diagnostic tasks, the implementation of Mobilenet V2 is relatively simple. Results demonstrate that the suggested model, having a general accuracy of 96.58%, a sensitivity of 95.17%, and a specificity of 98.58%, gives better predictions than separate models for classifying leukemia.

Index Terms - ALL; Deep learning; CNN; Medical imaging; Mobilenet V2;

I. INTRODUCTION

The most prevalent cancer type is leukemia among all blood cancers across all age categories, particularly in youngsters. The excessive expansion and inadequate blood cell growth, which can injure bone marrow, red blood cells, and the immune system, are to blame for this aberrant condition. Hematologists at cell transplant facilities can distinguish between various kinds of leukemia and provide a diagnosis based on microscopic images. Leukemia types can be identified more readily than others if the slide is appropriately stained, but more tools are required to identify underlying leukemia. A leukemia diagnosis is challenging due to the early stages minimal symptoms, which has always been a problem for researchers, physicians, and hematologists.

The majority of conventional computer-aided structures use machine learning and image processing methods, which typically require certain steps, such as segmentation, pre-processing, classification, and feature extraction. But each step's success is dependent upon the success of the one before it. As an illustration, success in feature extraction is a prerequisite for classification, which in turn depends on segmentation. Therefore, getting high classification accuracy requires completing all phases, each of which is challenging and has a particular challenge.

In the United States, there are 5930 new instances of Acute Lymphoblastic Leukemia (ALL) diagnosed in 2018, and 1500 individuals, including both adults and children, are anticipated to pass away from ALL. The chance of acquiring Acute Lymphoblastic Leukemia is moderately greater in men than women and lower in black Americans than whites. However, early detection of cancer will improve patient survival rates because it is very resistant to treatment. Given the size of histopathological images, evaluating them in a truly typical way is frequently difficult, unreliable, and time-consuming because some images have morphologies that are incredibly varied and difficult to analyze. Therefore, For effective early treatment, it is necessary to develop accurate and trustworthy methods for cancer detection. According to the findings of several studies, as process capabilities progress, hidden trends, patterns, and linkages are frequently found by utilizing the application of information mining methodologies in a variety of different fields [1]–[4]. Examples of healthy cells and ALL describes in Fig 1.

This study proposes a high-accuracy classification algorithm based on transfer learning that can distinguish between abnormal and healthy blood smear images. The major contributions of the authors are listed as follows-

- A new type of convolutional layer called Depthwise Separable convolution is used in Mobilenet V2, a CNN architecture that is substantially quicker and smaller. The Mobilenet V2 models are significantly faster than the Mobilenet V1. It is around 30–40% faster, requires 30% fewer parameters, utilises 2 times less procedures, and has superior precision.
- Transfer learning incorporates resource conservation and increased effectiveness when training new models. It can also help with model training when there are just unlabelled datasets available because the majority of the model will already have been trained. Additionally, it shortens training time.

Rest of the paper is organized as following: Section II discusses the related studies, materials and methods are discussed in section III whereas experiments and results are presented in section IV and the paper is concluded in section V.

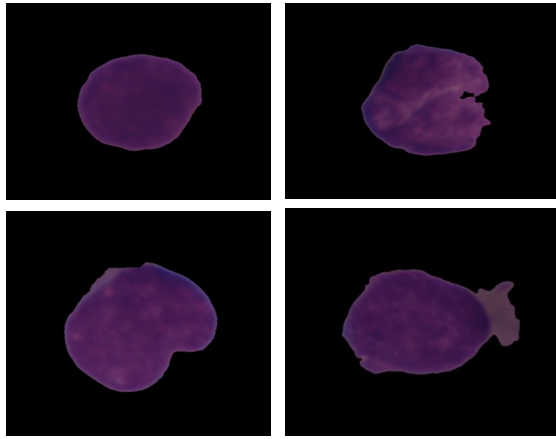


Fig. 1: Acute Lymphoblastic Leukemia (ALL) cells.

II. RELATED STUDIES

Over the years, in the literature, some methods for automatically detecting leukemia on microscopic pictures have been documented. Singhal et al. [5], On the basis of geometric data and local binary pattern (LBP) texture features, an SVM classifier has been utilized to automatically predict acute lymphoblastic leukemia. Compared to the shape features' accuracy, which is only 88.79%, the LBP texture features' accuracy is higher. According to experimental results which are 89.72%. The suggested model by Yu et al. [6], combines InceptionV3, VGG16, VGG19, ResNet50, and Xception with progressive convolution neural networks to develop a convolutional neural network-based system for automatic cell recognition. The projected model's outcomes are compared to those of legacy machine learning techniques like Logistic Regression (LR), Decision Tree (DT), Support Vector Machine (SVM), and K-Nearest Neighbors (KNN). This strategy using CNN models produced an accuracy of 88.50%.

In the suggested model by Mohamed et al. [7], Every image's color space is transformed into YCbCr in order to create the gaussian distribution of the values of Cr and Cb. Various features like texture, size, and morphology are computed in order to train a classifier. When using the Random Forest classifier to diagnose leukemia (AML and ALL) and myeloma, the suggested model has a 94.3% accuracy rate. Mohapatra et al. [8], introduced a method for image identification of acute leukemia of the bone marrow and stained blood smears. In order to train an ensemble model, features were taken from the input photos. The results viewed that in comparison to other common classifiers, like multilayer perceptron (MLP), radial basis function network (RBFN), K-nearest neighbor, naive Bayesian (NB), and SVM, the classifiers' collective average prediction accuracy was 94.73%, and their average specificity and sensitivity were also

above 90%.

In the suggested model by Patel et al. [9], K-mean clustering was used to simulate the detection of leukemia. The model could also figure out how much of an infection with leukemia, there was in microscopic pictures. The Patel approach had an accuracy rate of 93.57%.

Suggested the model, Mourya et al. [10], proposed using two CNN architectures in a deep learning-based hybrid architecture to increase classification accuracy. The model demonstrated 89.70% accuracy when tested on 636 samples of ALL and normal cells.

Yasmin et al. [11], The capture of microscopic pictures of blood samples is where the proposed framework begins. As additional data is needed for deep neural networks (DNN) training and higher performance, the problem of limited data is then resolved via data augmentation methods. To diagnose leukemia, a deep CNN architecture-based Squeeze and excitation learning are suggested. The proposed model's accuracy is 94%.

Das et al. [12], designed an automatic Covid-19 diagnosis model for identification of the disease, by analyzing image of their chest X-ray. The images classified into 3 categories as - Covid negative, Covid positive, other pneumonia. The model is trained by employing three learning methods: CNN, ResNet-50, and VGG-16.

Finally, Kumar et al. [13], The suggested system is trained using deep learning to be able to differentiate between attack and benign network data. The UNSW-NB15 and Bot-IoT datasets are used to analyze the proposed approach, and the results show that the model is successful in classifying network traffic as attack or benign.

III. MATERIALS AND METHODS

A. Methodology

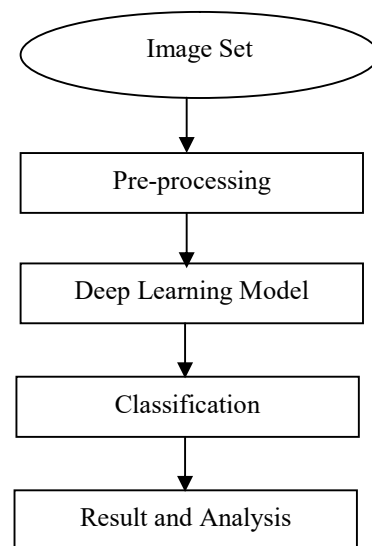


Fig. 2: Block Diagram of the Proposed Method

Our method is broken down into the following steps: first, we use various augmentation approaches and pre-processing for improving the visual information of each input photo in order to make important structures more visible. Once the input photos are ready, they are employed in the proposed transfer learning architecture's feature extraction stage. For our model, we investigate deep learning architecture, specifically Mobilenet V2. Using 13 convolutions with 3 x 3 convolution filters, two 4096 fully-connected layers, max-pooling layers, and a softmax classifier are all part of the relatively simple yet highly powerful MOBILENETV2 design ImageNet database, which has a million photos in 1000 categories, is used to train mobile net. A multi-layer perceptron was trained to categorize each photo into its associated class probability using extracted features. Finally, the test images are used to gauge the proposed architecture's performance.

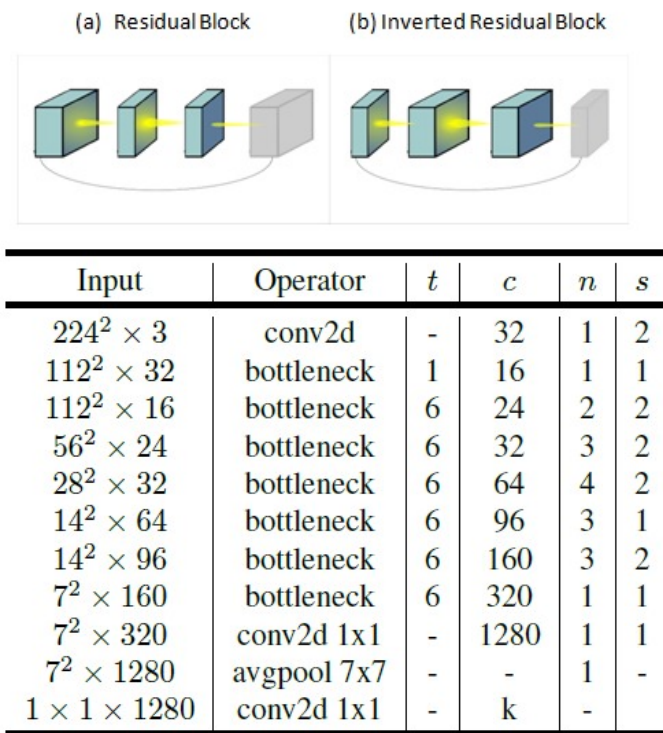


Fig 3: The overall architecture of Mobilenet V2

B. Experimental Dataset

The dataset employed in this work is based upon SBILab's publicly accessible as part of the ISBI 2019 challenge, classification of Normal over B-ALL White Blood Cancer Microscopic Images of Malignant Cells [14]. The photos are saved using a 24-bit RGB color scheme and have a resolution of 450×450 pixels. Each cell measures roughly 300×300 pixels in size.

Data Pre-processing

1) Normalization: In this experiment, more than one normalization technique is utilized to assess the efficiency of different techniques to normalize the input images, We begin by dividing the standard deviation of the mean RGB value across all photos in training set by the mean RGB value. as suggested in [15]. Additionally, as part of the pre-processing, we normalize the photos using ImageNet mean subtraction. The ImageNet mean comes from the ImageNet database and is a pre-calculated constant. [16].

2) Resizing: In order to verify that each cell is situated in the center and to eliminate the nearby background regions, we used bi-cubic interpolation to scale each image from the actual size image center of 450×450 pixels into a suitable size of 380×380 pixels as shown in Fig. 1.

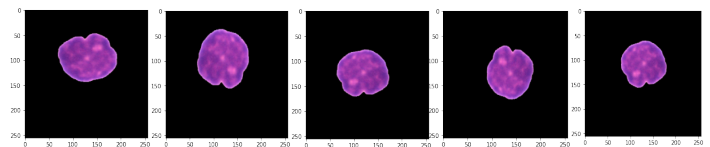


Fig 4: Image after preprocessing

3) Data Augmentation: CNN performed state-of-art across a variety of activities [17]–[19]. However, performance of CNN is significantly impacted by the volume of training data. Collecting good clinical photographs is difficult because of problems with data privacy in the medical industry. As recommended by recent studies, we have used a variety of data augmentation approaches to maximize CNN's performance to deal with the problem of restricted dataset size and prevent over-fitting issues [20], [21] featuring brightness correction, flipping the horizontal and vertical axis, and intensity changes. Table I displays the dataset's class distributions both after and before data augmentation.

TABLE I: Class distributions overall, both before and after data augmentation

Cell type	Number of images	
	Before augmentation	After augmentation
ALL cells	7272	53591
Healthy cells	3389	27930

C. Mobilenet V2 Architecture

The hybrid CNN model is proposed that mixes low-level characteristics from middle layers to create high-level feature maps for discrimination for identifying immature leukemic blasts. This method makes use of two known CNN architectures, VGG16 and VGG19, which have illustrated significant performance in different tasks using computer vision [22], [23]. The initial weights for the VGG16 architecture are generated from weights obtained through transfer learning from MobileNetV2. As demonstrated in

Figure 2, Features are obtained from five convolutional layers from the VGG19 architecture. Then an average pooling layer was placed over each of them. We have combined all to transform a single feature vector after that. Thereafter, using the ReLU activation function, we connect 256 hidden units in a new fully connected (FC) layer. Finally, at the classifier layer, Softmax non-linearity activation functions are present in two output neurons that are associated with normal and malignant cases. In addition to the already extracted deep features, these features that were retrieved from specific intermediary layers can be used in addition to gaining highly discriminative skills. This method yields higher accuracy with a lower mistake rate and the identification of more complicated designs from every input photo. When using very deep architecture to train small samples, subpar local minima and vanishing gradients may be problems. Using a global average pooling layer reduces the number of parameters in highly deep architecture, which is its main advantage.

Model: "mobilenetv2_1.00_224"

Layer (type)	Output Shape	Param #	Connected to
input_2 (InputLayer)	(None, None, None, 3)	0	[]
Conv1 (Conv2D)	(None, None, None, 32)	864	['input_2[0][0]']
bn_Conv1 (BatchNormalization)	(None, None, None, 32)	128	['Conv1[0][0]']
Conv1_relu (ReLU)	(None, None, None, 32)	0	['bn_Conv1[0][0]']
Conv_1_bn (BatchNormalization)	(None, None, None, 1280)	5120	['Conv1[0][0]']
out_relu (ReLU)	(None, None, None, 1280)	0	['Conv_1_bn[0][0]']

Total params: 2,257,984
Trainable params: 2,223,872
Non-trainable params: 34,112

Fig. 5: Architecture of the Mobilenet V2.

D. Evaluation Metrics

To evaluate the effectiveness of the advised course of action—accuracy, sensitivity, and specificity—three regularly used evaluation metrics—are taken into consideration. The degree of overall correctness is represented by accuracy, which is determined by dividing the whole number of test images by the number of correctly identified ALL instances.

$$\text{Accuracy}(\%) = (TP + TN / TP + TN + FP + FN) \times 100 \quad (1)$$

The ratio of true positive results to all other actual positives in illness diagnosis is called sensitivity, often known as the True Positive Rate (TPR). The model needs to be sensitive if there aren't many cancer samples in the dataset given.

$$\text{Sensitivity}(\%) = (TP / TP + FN) \times 100 \quad (2)$$

The true ratio of negative findings to all actual negatives is measured by specificity, often known as the True Negative Rate (TNR). As a result of its high specificity, the model is effective in identifying healthy cases.

$$\text{Specificity}(\%) = (TN / TN + FP) \times 100 \quad (3)$$

IV. EXPERIMENT AND RESULTS

A. Experimental Setup

The training set contains 70% of the photos from each class, for our experiment, 20% are from the validation set, and 10% are from the test set. For Adam, 0.001 was chosen as the learning rate. To avoid overfitting, in the fully linked layer, we applied the ReLU activation function with a dropout rate of 0.4. The size of the batch was set to 32 to fit within the GPU RAM. Every model undergoes 100 training iterations. Our experiment runs on Google Colab with GPU and is put into practice in Python using the Keras module and the Tensorflow backend deep learning framework.

B. Results and Discussion

The 967 test samples from the ISBI 2019 challenge that are not even utilized during training are where the acquired results came from. These test sets include 312 typical instances and 655 ALL cases. The effectiveness of each architecture is assessed separately in order to support the performance of the suggested method. Table I compares the various Mobilenet, VGG16, and VGG19 architectures with the suggested model. Table I shows that, on the presented dataset, our suggested technique performs noticeably better than the individual structures. Our model significantly increases the accuracy of VGG16 by up to 16%, VGG19 by up to 14%, and MobileNet by 8.58%.

TABLE II: The suggested model and simple pre-trained networks produce classification results. Bold values show the best result.

	<i>Accuracy</i>	<i>Sensitivity</i>	<i>Specificity</i>
Plain MobileNet	88.00	86.66	92.24
Plain VGG16	80.77	78.21	96.32
Plain VGG19	82.74	79.58	92.36
Proposed model	96.58	95.17	98.58

In Table II, our suggested ensemble is contrasted with a few recent research from the literature for the sake of comparison. According to Table III, the suggested approach performs more accurately when compared to other studies.

TABLE III: The comparison of the proposed method with current studies on the detection of leukemia

Dataset	Method	Accuracy	Year
ALL-IDB2	Singhal et al. [5]	89.720%	2014
DTH	Yu et al. [6]	88.500%	2017
MISP	Mohamed et al. [7]	93.000%	2018
IGH	Mohapatra et al. [8]	94.730%	2013
ALL-IDB	Patel et al. [9]	93.750%	2015
ISBI	Mourya et al. [10]	89.620%	2018
ISBI	Bukhari et. al. [11]	94.000%	2022
ISBI	Proposed Approach	96.580%	

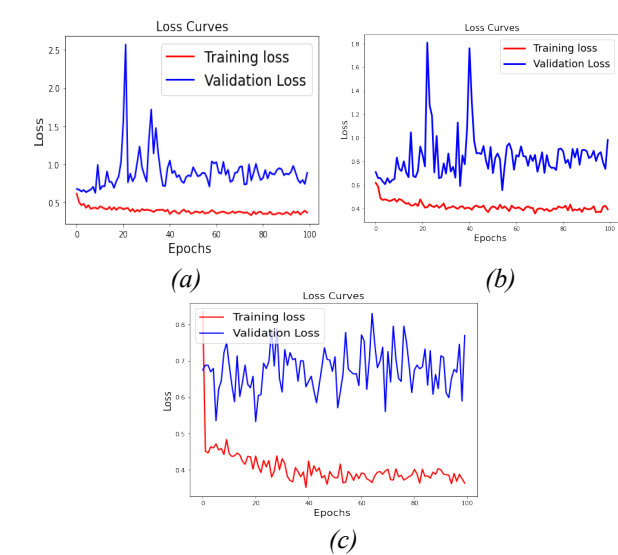


Fig 6: Loss curve of different models (a) VGG16 (b) VGG19 (c) Proposed Model

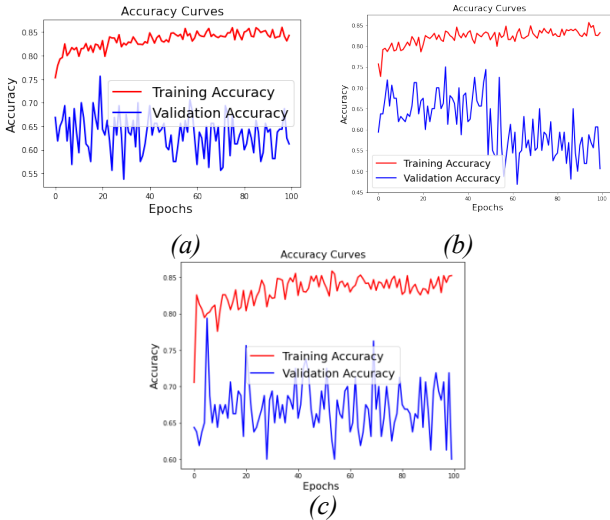


Fig 7: Accuracy curve of different models (a) VGG16 (b) VGG19 (c) Proposed Model

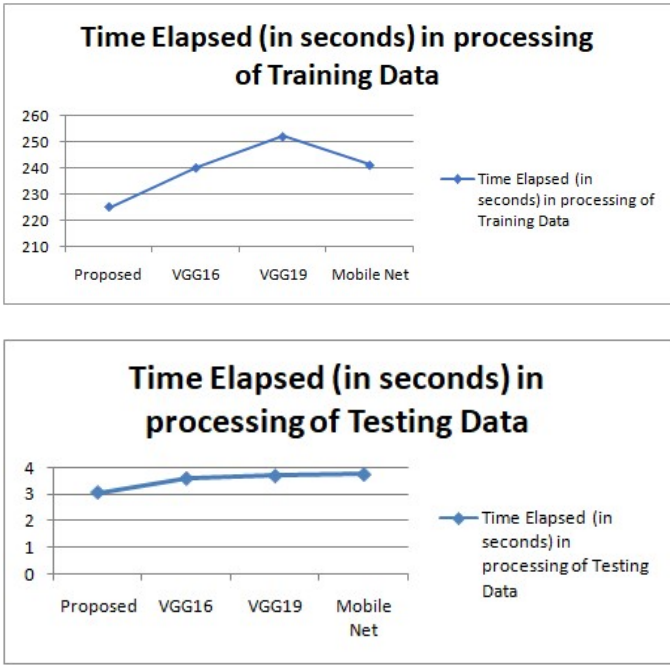


Fig 8: Performance analysis based on time complexity

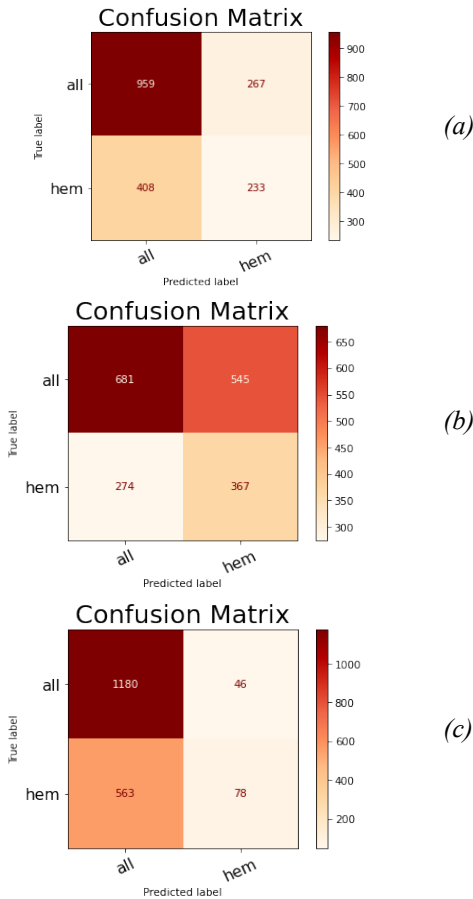


Fig 9: Confusion Matrix of different models (a) VGG16 (b) VGG19 (c) Proposed Model

V. CONCLUSION

For the automatic classification of healthy cells and ALL using Mobilenet V2, a new method is presented to identify features from various levels of abstraction, one well-known CNN, called MobilenetV2, is used. By using fewer parameters, this method not only aids in the learning of more complicated patterns but also deals with the issues of disappearing subpar local minima and gradients. According to the results, merging deep model learned features enhances performance and produces a result that is more accurate (96.58%) than using individual state-of-the-art networks. In order to observe the change in accuracy for future research areas, we plan to use Transfer learning of various CNN designs.

REFERENCES

- [1] S. H. Kassani, "Introducing a hybrid model of DEA and data mining in evaluating efficiency. Case study: Bank Branches," Apr. 2015.
- [2] E.-H. A. Rady and A. S. Anwar, "Prediction of kidney disease stages using data mining algorithms," *Inform. Med. Unlocked*, vol. 15, p. 100178, Jan. 2019, doi: 10.1016/j.imu.2019.100178.
- [3] M. C. Thomas, W. Zhu, and J. A. Romagnoli, "Data mining and clustering in chemical process databases for monitoring and knowledge discovery," *J. Process Control*, vol. 67, pp. 160–175, Jul. 2018, doi: 10.1016/j.jprocont.2017.02.006.
- [4] S. Mardanisamani *et al.*, "Crop Lodging Prediction from UAV-Acquired Images of Wheat and Canola using a DCNN Augmented with Handcrafted Texture Features," arXiv, Jun. 18, 2019, doi: 10.48550/arXiv.1906.07771.
- [5] V. Singhal and P. Singh, "Local Binary Pattern for automatic detection of Acute Lymphoblastic Leukemia," in *2014 Twentieth National Conference on Communications (NCC)*, Feb. 2014, pp. 1–5, doi: 10.1109/NCC.2014.6811261.
- [6] W. Yu *et al.*, "Automatic classification of leukocytes using deep neural network," in *2017 IEEE 12th International Conference on ASIC (ASICON)*, Oct. 2017, pp. 1041–1044, doi: 10.1109/ASICON.2017.8252657.
- [7] H. Mohamed *et al.*, "Automated detection of white blood cells cancer diseases," in *2018 First International Workshop on Deep and Representation Learning (IWDRL)*, Mar. 2018, pp. 48–54, doi: 10.1109/IWDRL.2018.8358214.
- [8] S. Mohapatra, D. Patra, and S. Satpathy, "An ensemble classifier system for early diagnosis of acute lymphoblastic leukemia in blood microscopic images," *Neural Comput. Appl.*, vol. 24, no. 7, pp. 1887–1904, Jun. 2014, doi: 10.1007/s00521-013-1438-3.
- [9] N. Patel and A. Mishra, "Automated Leukaemia Detection Using Microscopic Images," *Procedia Comput. Sci.*, vol. 58, pp. 635–642, Jan. 2015, doi: 10.1016/j.procs.2015.08.082.
- [10] S. Mourya, S. Kant, P. Kumar, A. Gupta, and R. Gupta, "LeukoNet: DCT-based CNN architecture for the classification of normal versus Leukemic blasts in B-ALL Cancer," arXiv, Nov. 04, 2018, doi: 10.48550/arXiv.1810.07961.
- [11] M. Bukhari, S. Yasmin, S. Sammad, and A. A. Abd El-Latif, "A Deep Learning Framework for Leukemia Cancer Detection in Microscopic Blood Samples Using Squeeze and Excitation Learning," *Math. Probl. Eng.*, vol. 2022, p. e2801227, Jan. 2022, doi: 10.1155/2022/2801227.
- [12] A. K. Das, S. Kalam, C. Kumar, and D. Sinha, "TLCoV- An automated Covid-19 screening model using Transfer Learning from chest X-ray images," *Chaos, Solitons & Fractals*, vol. 144, p. 110713, Mar. 2021, doi: 10.1016/j.chaos.2021.110713.
- [13] V. Kumar, S. Kalam, A. Das, and D. Sinha, "Attack Detection Scheme Using Deep Learning Approach for IoT," 2021, pp. 17–30, doi: 10.1007/978-981-16-4294-4_2.
- [14] S. Mourya, S. Kant, P. Kumar, A. Gupta, and R. Gupta, "ALL Challenge dataset of ISBI 2019," The Cancer Imaging Archive, 2019, doi: 10.7937/TCIA.2019.DC64I46R.
- [15] Z. Yu, X. Jiang, T. Wang, and B. Lei, "Aggregating Deep Convolutional Features for Melanoma Recognition in Dermoscopy Images," Sep. 2017, pp. 238–246, doi: 10.1007/978-3-319-67389-9_28.
- [16] A. Krizhevsky, I. Sutskever, and G. E. Hinton, "ImageNet Classification with Deep Convolutional Neural Networks," in *Advances in Neural Information Processing Systems*, 2012, vol. 25, Accessed: Dec. 01, 2022. [Online]. Available: <https://papers.nips.cc/paper/2012/hash/c399862d3b9d6b76c8436e924a68c45b-Abstract.html>
- [17] J.-H. Choi and J.-S. Lee, "EmbraceNet: A robust deep learning architecture for multimodal classification," *Inf. Fusion*, vol. 51, pp. 259–270, Nov. 2019, doi: 10.1016/j.inffus.2019.02.010.
- [18] A. B. Oktay and A. Gurses, "Automatic detection, localization and segmentation of nano-particles with deep learning in microscopy images," *Micron*, vol. 120, pp. 113–119, May 2019, doi: 10.1016/j.micron.2019.02.009.
- [19] L. Li, X. Zhao, W. Lu, and S. Tan, "Deep learning for variational multimodality tumor segmentation in PET/CT," *Neurocomputing*, vol. 392, pp. 277–295, Jun. 2020, doi: 10.1016/j.neucom.2018.10.099.
- [20] S. Hosseinzadeh Kassani and P. Hosseinzadeh Kassani, "A comparative study of deep learning architectures on melanoma detection," *Tissue Cell*, vol. 58, pp. 76–83, Jun. 2019, doi: 10.1016/j.tice.2019.04.009.
- [21] T.-C. Pham, C.-M. Luong, M. Visani, and V.-D. Hoang, "Deep CNN and Data Augmentation for Skin Lesion Classification," in *Intelligent Information and Database Systems*, Cham, 2018, pp. 573–582, doi: 10.1007/978-3-319-75420-8_54.
- [22] M. Santin *et al.*, "Detecting abnormal thyroid cartilages on CT using deep learning," *Diagn. Interv. Imaging*, vol. 100, no. 4, pp. 251–257, Apr. 2019, doi: 10.1016/j.diii.2019.01.008.
- [23] S. F. Chevtchenko, R. F. Vale, V. Macario, and F. R. Cordeiro, "A convolutional neural network with feature fusion for real-time hand posture recognition," *Appl. Soft Comput.*, vol. 73, pp. 748–766, Dec. 2018, doi: 10.1016/j.asoc.2018.09.010.